OTA를 위한 Bootloader설계 및 SW Binary 위/변조 감지의 이해 프로젝트

2026.03.03 ~ 2026.03.06

요구사항

- **싱글 뱅크 구조의 메모리 MCU**에서 UDS 프로토콜을 이용한 **Reprogramming 도중 전원이 차단되거나 통신 오류가 발생하면 Application이 손상되어 Bootlader만 남아 5초 주기로 재부팅되는 문제**가 발생
- Reprogramming 대상 **바이너리 위/변조 방지 및 유출**을 위한 보안

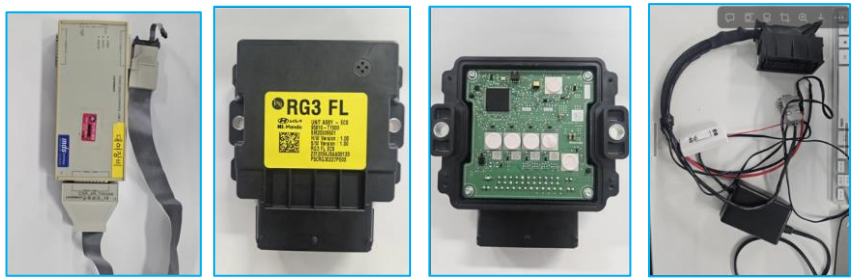
담당역할

- 개인 프로젝트

사용 Tool

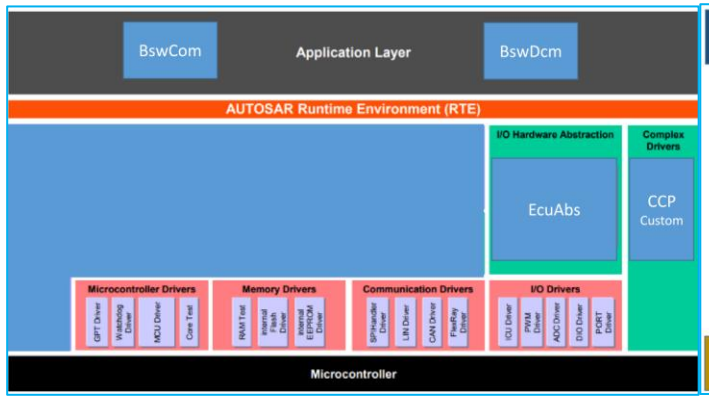
- IDE: Aurix Development Studio
- Vector XL 드라이버를 이용한 House Tool: Intelligent Can, Toolsus
- MCal Configuration: EB Tresos
- Trace32

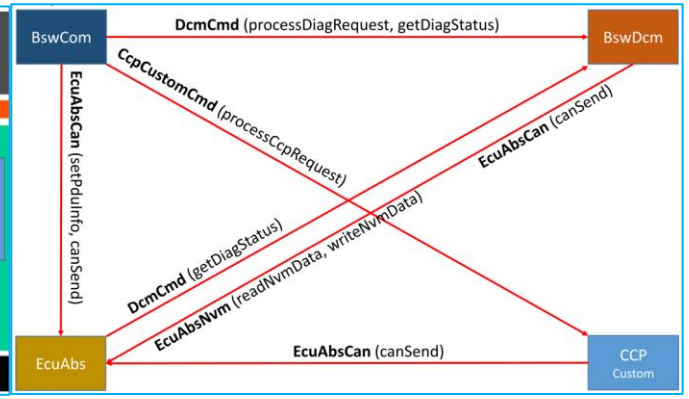
사용장비



Trace32 | 전자제어 Infineon TC234LP MCU | Wiring Harness

Project Architecture





결과

d:₩OTA_Training₩SHA_256_1₩main.exe

```
a5edde56e5cce1055e375022002c3260166624b1ee1ab9b18ddbf65b1a9b23a7
D:₩OTA_Training₩SHA_256_O>CertUtil -hashfile SCU_Gen6_Training.bin SHA256
SHA256의 SCU_Gen6_Training.bin 해시:
993a1300901b050b23765fda1c9cefbdae4afaa391ca935d0346d2d4d59c5ebf
CertUtil: -hashfile 명령이 성공적으로 완료되었습니다.
```

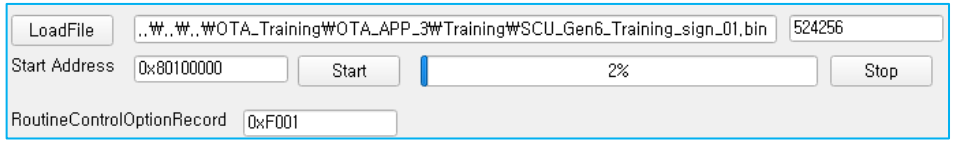
KISA의 **SHA256 해시함수 포팅을 활용한 서명 암호화**

78759	93.674209	0.100000	Rx	None	200	8		63 0B 00 00 00 00 00 00
78760	93.756342	0.499884	Tx	None	7D3	8	GST_ECS	04 31 01 00 01 55 55 55
78761	95.408456	2.144100	Rx	None	7DB	8	ECS_GST	04 71 01 00 01 AA AA AA
78762	95.408742	1.734533	Rx	None	200	8		64 0B 00 00 00 00 00 00
78763	95.501156	0.092414	Rx	None	200	8		65 0B 00 00 00 00 00 00
78764	95.517278	1.760936	Tx	None	7D3	8	GST_ECS	02 11 01 55 55 55 55 55
78765	95.521145	0.112689	Rx	None	7DB	8	ECS_GST	02 51 01 AA AA AA AA AA
78766	95.624143	0.122986	Rx	None	200	8		00 01 00 00 00 00 00 00

결과

(3-4Byte) 00: 현재 Application, 0B: Bootloader

40	3.641287	0.100024	Rx	None	200	8		5D 00 00 00 00 00 00 00
41	3.741286	0.100000	Rx	None	200	8		5E 00 00 00 00 00 00 00
42	3.746226	2.010538	Tx	None	7D3	8	GST_ECS	02 10 02 55 55 55 55 55
43	3.751256	2.009678	Rx	None	7DB	8	ECS_GST	06 50 02 00 32 01 F4 AA
44	3.853074	0.111788	Rx	None	200	8		00 0B 00 00 00 00 00 00
45	3.953074	0.100000	Rx	None	200	8		01 0B 00 00 00 00 00 00
1826	7.870169	0.000483	Tx	None	7D3	8	GST_ECS	2C 01 5E 14 DA 01 89 CF
1827	7.870407	0.000238	Tx	None	7D3	8	GST_ECS	2D 20 08 91 00 00 F7 D9
1828	7.969907	0.099975	Rx	None	200	8		27 0B 00 00 00 00 00 00
1829	8.069898	0.099992	Rx	None	200	8		28 0B 00 00 00 00 00 00
1876	12.769706	0.099992	Rx	None	200	8		57 0B 00 00 00 00 00 00
1877	12.869706	0.100000	Rx	None	200	8		58 0B 00 00 00 00 00 00
1878	12.982706	0.113000	Rx	None	200	8		00 00 00 00 00 00 00 00
1879	13.082706	0.100000	Rx	None	200	8		01 00 00 00 00 00 00 00



미사용 Flash Memory를 사용한
Reprogramming 중단 시 기존 App으로 복귀

SHA256 관련 Code 분산을 통한 RID_CheckProgrammingDependencies 응답 시간 **약 500ms 감소**

78750	92.681224	0.099975	Rx	None	200	8		69 0B 00 00 00 00 00 00
78751	92.687729	0.505070	Tx	None	7D3	8	GST_ECS	04 31 01 00 01 55 55 55
78752	93.850370	1.658913	Rx	None	7DB	8	ECS_GST	04 71 01 00 01 AA AA AA
78753	93.938164	1.256940	Rx	None	200	8		6A 0B 00 00 00 00 00 00
78754	93.949805	1.262076	Tx	None	7D3	8	GST_ECS	02 11 01 55 55 55 55 55
78755	93.958177	0.107807	Rx	None	7DB	8	ECS_GST	02 51 01 AA AA AA AA AA
78756	94.061199	0.123036	Rx	None	200	8		00 01 00 00 00 00 00 00